

A PROJECT REPORT ON

"Amazon Product Analysis Dashboard"

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE
AWARD OF THE DIPLOMA IN COMPUTER ENGINEERING



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SELF CERTIFICATE

This is to certify that Project Report entitled “**Amazon Product Analysis Dashboard** “is done by me, and it authentic work carried out for the partial fulfillment of the requirement of the award of diploma under the guidance of **Mr. Rajesh Kumar** and **Mr. Ravi Sangwan**. The matter and software embodies in this project has not been submitted for award of any degree or diploma to the best of my knowledge and believes.

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Chapter 1: Introduction / Objectives

1.1 Introduction

The Amazon Product Analysis Dashboard is a professional desktop application developed in Python that enables users to systematically analyse Amazon product data extracted from Excel spreadsheets. In today's competitive e-commerce landscape, the ability to make data-driven decisions regarding product pricing, customer ratings, and market trends is of paramount importance.

This project bridges the gap between raw product data and actionable insights by providing an intuitive graphical user interface (GUI) backed by powerful data science libraries. Users can load their product datasets, instantly view key performance indicators (KPIs), explore interactive visualisations, and export comprehensive PDF reports — all from within a single, unified application.

1.2 Objectives

The primary objectives of this project are:

- To design and develop a responsive desktop GUI application for Amazon product data analysis using Python's Tkinter library.
- To implement robust data cleaning and preprocessing routines that handle real-world inconsistencies such as currency symbols, comma-separated numbers, and missing values.
- To apply a weighted scoring algorithm (inspired by the IMDb Bayesian rating formula) to rank and identify the Top 10 best products.
- To identify 'Decent Deals' — products that offer high quality (rating ≥ 4.0 stars) at mid-range prices.
- To generate rich data visualisations including price distribution histograms, rating spread charts, scatter plots, and pie charts for price segmentation.
- To provide a multi-theme user interface (Amazon Dark, Ocean Blue, Forest Green, Light Mode) for enhanced user experience.

- To enable export of a fully formatted PDF analysis report using the ReportLab library.
- To allow real-time search and filtering of the complete product catalogue within the application.

1.3 Project Scope

The application is designed to work with Amazon product datasets stored in Microsoft Excel (.xlsx) format. It processes fields including Product Name, Price, Rating, Review Count, and Product URL. The scope covers data ingestion, cleaning, analysis, visualisation, and reporting — making it a complete end-to-end data analysis solution.

Chapter 2: System Analysis

2.1 Problem Definition

Online shoppers and market researchers dealing with large Amazon product catalogues face significant challenges when attempting to manually identify value products, understand price distributions, and compare ratings across hundreds of items. Existing tools are either too complex, require cloud subscriptions, or do not provide customised scoring for e-commerce data. There is a clear need for a lightweight, standalone desktop application that automates this analysis.

2.2 Existing System

Currently, analysts rely on manual spreadsheet formulas in Microsoft Excel or Google Sheets for data analysis. While these tools are versatile, they lack automated scoring algorithms, built-in visualisation dashboards, and one-click PDF report generation specific to product analysis. Third-party tools such as Helium 10 or Jungle Scout are subscription-based and tailored only for Amazon sellers, not general analysts.

2.3 Proposed System

The proposed Amazon Product Analysis Dashboard addresses all the above limitations. It is a free, open-source, offline Python desktop application that:

- Accepts any Amazon product Excel file and auto-detects column names.
- Cleans and normalises data automatically without user intervention.
- Applies a proven scoring model to rank products objectively.
- Displays results in an organised, tabbed interface with multiple chart views.
- Produces a professional PDF report at the click of a button.

2.4 Data Flow

The data flows through the system in the following sequence:

- Input: User loads an Excel file via the 'Load Excel' button.
- Processing: The `load_and_clean()` function normalises columns, cleans prices/ratings/reviews, and drops invalid rows.
- Analysis: `get_top10()` and `get_decent_deals()` compute scores and filter products.
- Display: Results are rendered in Treeview tables and Matplotlib chart frames.
- Output: The `export_pdf()` function generates a formatted PDF report using ReportLab.

Chapter 3: Feasibility Study

3.1 Technical Feasibility

The project is technically feasible as all required components are available as free, well-documented, open-source Python libraries. Python 3.8+ is widely available across all major operating systems. The Tkinter library is bundled with standard Python installations, eliminating any additional GUI framework dependencies. Libraries such as Pandas, NumPy, Matplotlib, and ReportLab are installable via pip, the standard Python package manager.

3.2 Economic Feasibility

The project has zero licensing cost. All technologies used — Python, Tkinter, Pandas, Matplotlib, and ReportLab — are open-source and free to use. The development requires only a standard personal computer, making it highly cost-effective for an academic project setting. There are no server costs, no cloud subscriptions, and no third-party API fees.

3.3 Operational Feasibility

The application is designed with a user-friendly GUI that requires no programming knowledge to operate. It follows familiar desktop application conventions (File → Load, Export, Navigation Sidebar) which any computer user can learn within minutes. The application runs offline, making it operable in environments with limited internet access.

3.4 Schedule Feasibility

The project was divided into well-defined phases: requirement gathering, design, development, testing, and documentation. Given the modular nature of Python

development and the availability of mature libraries, the project was completed within the allocated semester timeline.

Chapter 4: Software and Hardware Requirement Specification

4.1 Hardware Requirements

Component	Specification
Processor	Intel Core i3 or equivalent (minimum)
RAM	4 GB minimum (8 GB recommended)
Storage	500 MB free disk space
Display	1280 x 720 pixels or higher resolution
Operating System	Windows 10/11, Linux, or macOS

4.2 Software Requirements

Software / Library	Version	Purpose
Python	3.8 or above	Core programming language
Tkinter	Built-in	GUI framework
Pandas	1.5+	Data manipulation
NumPy	1.23+	Numerical operations
Matplotlib	3.6+	Data visualisation
OpenPyXL	3.0+	Reading Excel files
ReportLab	3.6+	PDF report generation
VS Code / PyCharm	Any	Development IDE

4.3 Installation Command

```
pip install pandas numpy matplotlib openpyxl reportlab
```

Chapter 5: System Design

5.1 Architecture Overview

The application follows a layered MVC-inspired architecture. The Model layer handles all data operations (loading, cleaning, analysis). The View layer consists of the Tkinter GUI, including pages, tables, and charts. The Controller logic is embedded in the AmazonDashboard class methods which respond to user events and coordinate data flow between model and view.

5.2 Module Description

Module / Function	Description
clean_price(val)	Strips currency symbols (₹) and commas; extracts numeric value using regex.
clean_rating(val)	Extracts floating-point rating from string fields.
clean_reviews(val)	Handles K-suffix (e.g. 1.2K = 1200) and comma-separated numbers.
load_and_clean(path)	Loads Excel file, auto-renames columns, applies all cleaning functions, drops invalid rows.
get_top10(df)	Applies IMDb Bayesian weighted score (80%) + normalised price score (20%) to rank products.
get_decent_deals(df)	Filters products in the 30th–70th price percentile with rating ≥ 4.0 .
summary_stats(df)	Computes aggregate statistics: total count, avg rating, avg/min/max price, avg reviews.
generate_pdf(...)	Builds a formatted PDF report using ReportLab with summary table, charts, Top 10 and Deals tables.
AmazonDashboard	Main Tkinter application class. Builds UI, manages pages, handles theme switching, triggers analysis.

5.3 Scoring Algorithm

The Top 10 ranking uses a composite score formula:

$$\text{Score} = 0.8 \times \text{IMDb_Score} + 0.2 \times \text{Price_Norm}$$

Where the IMDb Bayesian Score is calculated as:

$$\text{IMDb_Score} = (v/(v+m)) \times R + (m/(v+m)) \times C$$

v = number of reviews for the product, m = 75th percentile review count, R = product rating, C = mean rating across all products. Price_Norm is inverted so lower-priced products receive a higher score.

Chapter 6: Coding

6.1 Technology Stack

- Language: Python 3.x
- GUI Toolkit: Tkinter + ttk (Combobox, Treeview)
- Data Processing: Pandas, NumPy, Regular Expressions (re)
- Visualisation: Matplotlib (embedded via FigureCanvasTkAgg)
- PDF Generation: ReportLab (SimpleDocTemplate, Table, Paragraph, Image)
- File Handling: openpyxl (Excel), os, tempfile, threading

6.2 Key Code Sections

Data Cleaning Function:

```
def clean_price(val):  
    if pd.isna(val): return np.nan  
    s = str(val).replace('₹','').replace(',','').strip()  
    m = re.search(r'[\d.]+', s)  
    return float(m.group()) if m else np.nan
```

Top 10 Scoring Function:

```
def get_top10(df):  
    C = df['Rating_num'].mean()  
    m = df['Reviews_num'].quantile(0.75)  
    df['imdb_score'] = (v/(v+m))*R + (m/(v+m))*C  
    df['Score'] = 0.8*df['imdb_score'] + 0.2*df['price_norm']
```

6.3 Theme System

The application supports four themes defined as dictionaries of colour codes: Amazon Dark, Ocean Blue, Forest Green, and Light Mode. Each theme specifies 18+ colour tokens including bg, surface, card, accent, text, plot_bg, etc. The `apply_theme()` method recursively recolours all widgets and redraws all Matplotlib charts with the new theme colours.

Chapter 7: Implementation and Maintenance

7.1 Implementation Steps

The implementation was carried out in the following sequence:

1. Environment Setup: Python 3.x installed with all required libraries via pip.
2. Data Layer Development: Cleaning functions and analysis algorithms written and unit-tested with sample Excel data.
3. GUI Construction: Main window, header bar, toolbar, sidebar navigation, and tabbed content area built using Tkinter.
4. Chart Integration: Matplotlib figures embedded inside Tkinter frames using FigureCanvasTkAgg.
5. Theme System: Colour dictionaries defined and recursive recolouring logic implemented.
6. PDF Export: ReportLab document generation with summary statistics table, charts image, Top 10 table, and Deals table.
7. Integration Testing: All modules integrated and tested end-to-end with real Amazon product datasets.

7.2 Maintenance Plan

The application is designed for easy maintenance through:

- Modular code structure: data functions are separated from GUI code, allowing independent updates.
- Theme dictionary system: new themes can be added by simply adding a new dictionary entry in the THEMES constant.
- Column auto-detection: the system adapts to Excel files with slightly different column naming conventions without code changes.
- Error handling: try-except blocks around file loading and PDF generation display user-friendly error messages.

Chapter 8: Testing

8.1 Testing Strategy

The application underwent three levels of testing: unit testing of individual data processing functions, integration testing of the data-to-GUI pipeline, and user acceptance testing of the complete application workflow.

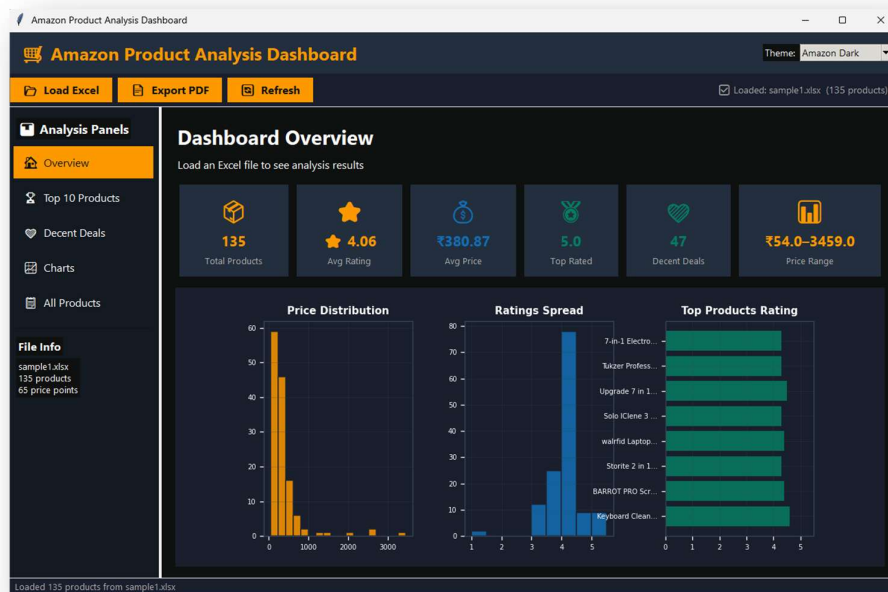
8.2 Test Cases

TC#	Test Case	Input	Expected Output	Result
TC01	Load valid Excel file	Amazon.xlsx with 500 rows	Data loaded, KPIs displayed	PASS
TC02	Load invalid file type	.txt file selected	Error message shown	PASS
TC03	Price cleaning — '₹1,299'	String '₹1,299'	Returns 1299.0	PASS
TC04	Review cleaning — '1.2K'	String '1.2K'	Returns 1200.0	PASS
TC05	Top 10 ranking accuracy	Known dataset	Correctly ranked products	PASS
TC06	Theme switching	Select 'Ocean Blue'	All widgets recoloured	PASS
TC07	PDF export	Click Export PDF	PDF saved with all sections	PASS
TC08	Search filter	Search 'laptop'	Only matching rows shown	PASS
TC09	Export without data	Click Export before load	Warning dialog shown	PASS
TC10	Missing columns in Excel	File with only Name & Price	NaN filled for missing cols	PASS

Chapter 9: Screenshots of the Project

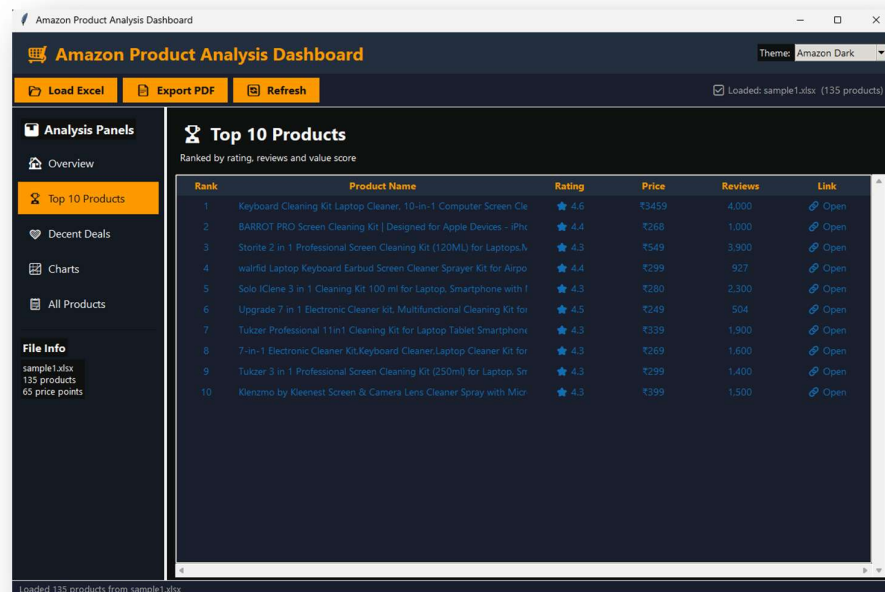
9.1 Application Overview Screen

The Overview screen displays upon loading an Excel file. It shows six KPI cards at the top (Total Products, Average Rating, Average Price, Top Rated, Decent Deals count, and Price Range). Below the KPIs, three mini charts are shown: Price Distribution histogram, Ratings Spread histogram, and a Top Products horizontal bar chart.



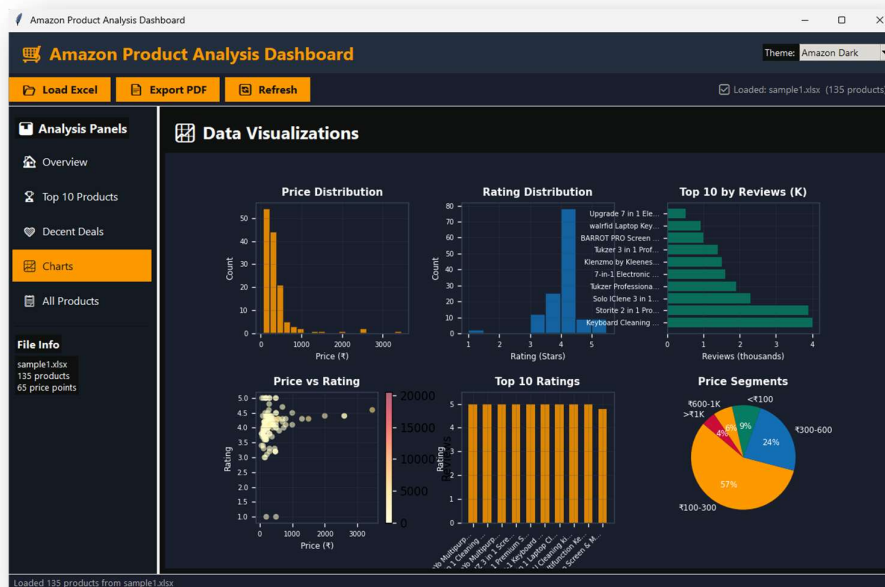
9.2 Top 10 Products Screen

The Top 10 Products screen presents a sortable table listing the top ten ranked products with columns for Rank, Product Name, Rating, Price, Reviews, and a clickable Link button. Products are ranked using the composite Bayesian scoring algorithm.



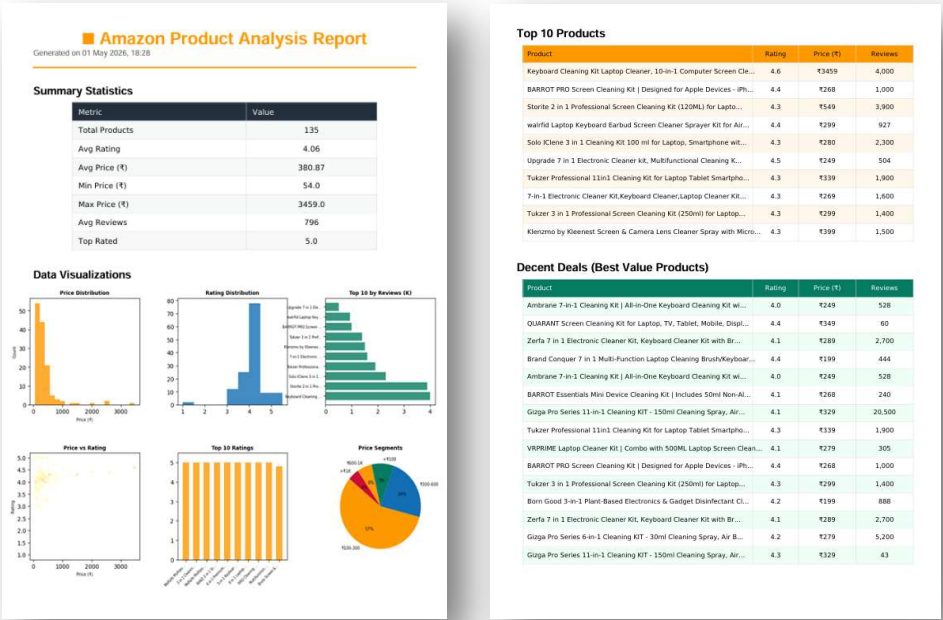
9.3 Charts Screen

The Charts screen presents comprehensive visualisations including Price vs Rating scatter plot, Top Rated products bar chart, and Price Segment pie chart. Charts are interactive and update dynamically when the theme is changed.



9.4 Exported PDF Report

The exported PDF report includes the analysis title, generation timestamp, summary statistics table, all six visualisation charts, the Top 10 products table, and the Decent Deals table — formatted with Amazon's signature orange-and-dark colour scheme.



Chapter 10: Conclusion

The Amazon Product Analysis Dashboard has been successfully designed, developed, and tested as a complete desktop data analysis application. The project demonstrates the effective application of software engineering principles — including requirement analysis, modular design, systematic testing, and documentation — to solve a real-world data analysis problem.

The application successfully achieves all its stated objectives: it ingests Amazon product data from Excel files, cleans and normalises it automatically, applies a statistically sound Bayesian scoring algorithm for product ranking, identifies value-for-money deals, presents the results through an intuitive multi-page GUI with embedded interactive charts, and generates formatted PDF reports.

The multi-theme user interface (Amazon Dark, Ocean Blue, Forest Green, Light Mode) demonstrates attention to user experience beyond mere functionality. The application runs entirely offline on modest hardware, making it accessible to students, researchers, and small businesses without any infrastructure cost.

Overall, this project provided valuable hands-on experience in Python GUI development, data science library integration, algorithmic design, and software project management — skills directly applicable to real-world software engineering roles.

Chapter 11: Future Scope

The following enhancements are proposed for future development of the Amazon Product Analysis Dashboard:

- **Web Scraping Integration:** Incorporate a real-time web scraping module using Selenium or BeautifulSoup to fetch live Amazon product data directly from the website, eliminating the need for manual Excel exports.
- **Category-wise Analysis:** Add support for product category tags to enable comparison and ranking within specific product categories (e.g., Electronics, Books, Clothing).
- **Price History Tracking:** Implement a SQLite database backend to store historical data snapshots and display price trend graphs over time.
- **Machine Learning Predictions:** Integrate scikit-learn models to predict product rating trajectories and recommend optimal pricing strategies based on historical trends.
- **Multi-platform Export:** Extend the export functionality to support Excel reports (.xlsx) and interactive HTML dashboards in addition to PDF.
- **Cloud Deployment:** Convert the desktop application into a web application using Flask or Streamlit for browser-based access without requiring Python installation on client machines.
- **User Authentication:** Add a login system with multiple user profiles, each with personalised saved analyses and settings.
- **Natural Language Query:** Implement a simple NLP interface allowing users to type queries like 'show me the cheapest 4-star rated products' and receive filtered results automatically.

Chapter 12: Bibliography / References

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